

Problem Statement & Data (19/20)

You provided a clear problem/purpose statement—to check if bike paths in NYC are heavily congested. It might have been strengthened if you connected bike congestion with larger issues like safety, tourism, reducing car traffic, etc. You made good use of data sources that were not part of our exercises in class.

Methodology (18/20)

Sampling makes sense when using large datasets. However, as we discussed, since your problem statement is focused on the volume of demand, you need to scale your sample back up. Otherwise, the sample represents a much smaller demand for bikes, unlikely to cause congestion in bike lanes. You adjusted the network ideas we used in class for biking (u-turns anywhere, no one ways, etc). Good use of the intersect (**split lines at point**) tool to turn bike paths into smaller units. Good use of **spatial join** with “share a line segment with” (**new operator**) to get count of segment uses.

Maps/Tables (18/20)

Good use of symbology to try and convey a lot of information. Size of points might be used to convey frequency of use.

Results/Interpretation (19/20)

It is really interesting that there are so many more stops than starts. This could be caused by sampling only at the end of the day when people might be returning bikes. There are a lot of questions that could have been asked for policy reasons, such as what percentage of trips begin and end on bike paths? If this percentage is high, the current bike path network might be sufficient. If this percentage is low, the current bike path network might need expansion and your analysis could suggest where. I like your idea to compare pre/post COVID pandemic rental levels.

Presentation (18/20)

A dedicated presentation might have made it easier to follow your work. But, you presented the major points well regardless.